

DAY 22: CORRELATION (PEARSON AUR SPEARMAN)

Advanced Data Analytics Masterclass Report

1. Pearson Linear Correlation Coefficient

Definition: Pearson Correlation Coefficient is a statistical measure that quantifies the strength and direction of a linear relationship between two continuous variables.

Pearson Correlation Coefficient ek statistical measure hai jo do continuous variables ke beech ke linear rishte ki mazbooti aur direction ko naapta hai.

It evaluates how close the data points are to a perfectly straight line, assuming that a change in one variable results in a proportional change in the other.

Yeh check karta hai ki data points ek perfectly seedhi line ke kitne paas hain, aur yeh maanta hai ki ek variable mein badlav se dusre variable mein bhi barabar ratio mein badlav aata hai.

Formula Explanation: The formula computes the ratio of the joint variance (covariance) of the two variables to the product of their individual standard deviations.

Formula dono variables ke aapas ke bhatkaw (covariance) ko unke alag-alag standard deviations ke multiplication se divide karke ratio nikalta hai.

$$r = \frac{\sum(X - \bar{X})(Y - \bar{Y})}{\sqrt{[\sum(X - \bar{X})^2 * \sum(Y - \bar{Y})^2]}}$$

$$r = \frac{\sum(X - X \text{ ka Mean})(Y - Y \text{ ka Mean})}{\sqrt{[\sum(X - X \text{ ka Mean})^2 * \sum(Y - Y \text{ ka Mean})^2]}}$$

1.1 Pearson Numerical Scenario 1: Uniform Gold Weight vs Selling Price

Scenario: A jewelry showroom tracks the gold weight in grams (X) and the final selling price in AED (Y) for premium items to audit pricing consistency.

Data Points: Item 1: X = 10g, Y = 3000 AED | Item 2: X = 20g, Y = 6000 AED | Item 3: X = 30g, Y = 9000 AED.

Step 1: Calculate the Mean (Average) of X and Y.

Mean of X ($\bar{X}$) = (10 + 20 + 30) / 3 = 20

Mean of Y ($\bar{Y}$) = (3000 + 6000 + 9000) / 3 = 6000

Step 2: Construct the Deviation and Product Table.

Item	X	Y	X - X̄	Y - Ȳ	(X-X̄)(Y-Ȳ)	(X-X̄)²	(Y-Ȳ)²
Item 1	10	3000	-10	-3000	30000	100	9000000
Item 2	20	6000	0	0	0	0	0
Item 3	30	9000	10	3000	30000	100	9000000
Sum (Σ)	60	18000	0	0	60000	200	18000000

Step 3: Apply the Values into the Pearson Formula.

$$\text{Denominator} = \sqrt{[\sum (X - \bar{X})^2 * \sum (Y - \bar{Y})^2]}$$

$$\text{Denominator} = \sqrt{[200 * 18,000,000]} = \sqrt{[3,600,000,000]} = 60,000$$

$$r = 60,000 / 60,000 = 1.0$$

Conclusion: The result shows a perfect positive linear relationship because the price scales exactly in proportion to the gold weight.

1.2 Pearson Numerical Scenario 2: Design Complexity vs Production Cost

Scenario: The operations manager analyzes how the design complexity score (X) impact the actual factory production cost in AED (Y).

Data Points: Batch 1: X = 50, Y = 2000 | Batch 2: X = 90, Y = 5000 | Batch 3: X = 70, Y = 1000 | Batch 4: X = 80, Y = 3000.

Step 1: Calculate Mean of X and Y.

$$\text{Mean of X } (\bar{X}) = (50 + 90 + 70 + 80) / 4 = 72.5$$

$$\text{Mean of Y } (\bar{Y}) = (2000 + 5000 + 1000 + 3000) / 4 = 2750$$

Step 2: Table Calculations.

Batch	X - X̄	Y - Ȳ	(X-X̄)(Y-Ȳ)	(X-X̄)²	(Y-Ȳ)²
Batch 1	-22.5	-750.0	16875	506.25	562500
Batch 2	17.5	2250.0	39375	306.25	5062500
Batch 3	-2.5	-1750.0	4375	6.25	3062500
Batch 4	7.5	250.0	1875	56.25	62500
Sum (Σ)	0	0	62500	875.0	8750000

Step 3: Final Calculation.

$$\text{Denominator} = \sqrt{[875.0 * 8,750,000]} = \sqrt{[7,656,250,000]} = 87,500$$

$$r = 62,500 / 87,500 = 0.714$$

Conclusion: The score is 0.714, indicating a strong positive correlation, but the deviation in Batch 3 prevents it from being a perfect 1.0 because the cost dropped significantly relative to complexity.

2. Spearman Rank Correlation Coefficient

Definition: Spearman Rank Correlation Coefficient is a non-parametric measure that evaluates the monotonic relationship between two ranked variables.

Spearman Rank Correlation Coefficient ek non-parametric measure hai jo do ranked variables ke beech ke monotonic rishte ka valuation karta hai.

It measures how well an arbitrary monotonic function can describe the relationship between two variables, without making assumptions about the frequency distribution of the data.

Yeh naapta hai ki koi bhi monotonic trend do variables ke rishte ko kitne acche se samjha sakta hai, bina data ke distribution ka koi assumption liye.

Formula Explanation: The formula focuses entirely on the differences between the ranks assigned to the values of each variable, instead of their actual numerical distances.

Formula asali numerical doori ko chhodkar, sirf har ek variable ki values ko diye gaye ranks ke beech ke farq (difference) par focus karta hai.

$$\rho = 1 - [6 * \sum d^2 / (N * (N^2 - 1))]$$

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2.1 Spearman Numerical Scenario 1: Showroom Experience vs Brand Trust Rank

Scenario: An audit is conducted across three branches to determine if more years of showroom operations (X) leads to a higher Brand Trust Score (Y).

Data Points: Branch 1: X = 2 years, Y = 10 | Branch 2: X = 5 years, Y = 50 | Branch 3: X = 10 years, Y = 60.

Step 1: Assign Ranks (Highest value gets Rank 1).

R_x (Experience Ranks): 10yrs = Rank 1, 5yrs = Rank 2, 2yrs = Rank 3

R_y (Trust Ranks): 60 Score = Rank 1, 50 Score = Rank 2, 10 Score = Rank 3

Step 2: Table of Rank Differences (d = R_x - R_y).

Branch	Rx (Experience Rank)	Ry (Trust Rank)	Diff (d)	Square (d²)
Branch 1	3	3	0	0
Branch 2	2	2	0	0
Branch 3	1	1	0	0
Sum (Σ)	-	-	0	0

Step 3: Compute Spearman Score (N = 3).

$$\rho = 1 - [6 * \sum d^2 / (N * (N^2 - 1))]$$

$$\rho = 1 - [6 * 0 / (3 * (3^2 - 1))] = 1 - 0 = 1.0$$

Conclusion: Spearman gives a perfect 1.0 because the direction of growth is perfectly aligned, even though the actual numeric growth steps are unequal.

2.2 Spearman Numerical Scenario 2: Design Complexity vs Making Charges

Scenario: Using the same jewelry dataset from section 1.2, we recalculate using Spearman to analyze positional rank trends.

Data Points: Item 1: X = 50, Y = 2000 | Item 2: X = 90, Y = 5000 | Item 3: X = 70, Y = 1000 | Item 4: X = 80, Y = 3000.

Step 1: Assign Ranks.

Rx (Design Ranks): 90 = Rank 1, 80 = Rank 2, 70 = Rank 3, 50 = Rank 4

Ry (Making Ranks): 5000 = Rank 1, 3000 = Rank 2, 2000 = Rank 3, 1000 = Rank 4

Step 2: Difference Calculations.

Item	Rx (Design Rank)	Ry (Making Rank)	Diff (d)	Square (d²)
Item 1	4	3	1	1
Item 2	1	1	0	0
Item 3	3	4	-1	1
Item 4	2	2	0	0
Sum (Σ)	-	-	0	2

Step 3: Apply Spearman Formula (N = 4).

$$\rho = 1 - [6 * \sum d^2 / (N * (N^2 - 1))]$$

$$\text{Numerator} = 6 * 2 = 12$$

$$\text{Denominator} = 4 * (4^2 - 1) = 4 * 15 = 60$$

$$\rho = 1 - (12 / 60) = 1 - 0.2 = 0.80$$

Conclusion: Spearman shows a score of 0.80, which is less sensitive to the massive scale drop in Item 3 compared to Pearson (0.714), as it only evaluated the shift in ranking order.